Revenue Maximization and Customer-centric Strategies for Mutual Fund Distribution

Final Report for the BDM Capstone Project

Submitted by

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1 Executive Summary:

M K Financial Services is a financial intermediary that helps its clients by providing services like mutual fund distribution, stock market broking, and insurance related services. This project focuses mainly on the mutual fund distribution business of the organization. A few problems the organization faces are no structured client acquisition strategies and lack of well-defined client categorization. Absence of forecasted growth strategies to trace for the organization results in variable growth rates over years. Determining periodic sales targets remain ignored because of organization's undivided focus on portfolio management.

The data collection focuses on two broad divisions: Client data which includes client profile as well as their investments over time; Time series data of AUM of the organisation and the number of clients every year. The client data focuses on analysing the client profiles and hence help in segmentation and categorisation of clients based on various profile features. It further helps in formulating acquisition and retention strategies. The time series data is instrumental in determining growth rates of AUM as well as number of clients. It can be further extended to determine targets for the organisation to achieve. This target setting provides a direction to the organisation for a measured revenue growth.

The results show the presence of clients who significantly contribute to the revenue of the organisation currently. Segmentation shows that the clientele can be clustered into meaningful groups with well-defined cluster characteristics. This is correctly verified by analysing the investment metrics across the clusters and the central tendencies of the numerical features among clusters remain well differentiated. The time series data was instrumental in determining growth rates. They were also able to explain the impact of external factors like micro and macro-economic conditions as well as market scenario. The growth rates were further used to forecast AUM for the organisation for the next 10 years and based on the forecasted AUM, targets were determined for the organisation in terms of new inflows yearly further dissecting into the number of clients needed to be acquired to keep up with the growth rates.

The interpretations show the importance of references in this business. A decent acquisition strategy for formulised involving the trend in salary earning individuals who have little to no responsibilities and are also single. These individuals have long term appreciation mindset through SIPs. As interpreted, SIPs are instrumental in boosting up the AUM continuously. The immediate call of action is to actively communicate with the top 70-80 AUM contributors and also start a structured acquisition process that involves leveraging references and forming a

network among salaried individuals. The targets were also determined for number of clients to be acquired. These show that the client growth should not be constant but follow a growth rate to achieve the forecasted AUM over the next 10 years.

2 Detailed Explanation of Analysis Process/Method

CLIENT_MASTER

1. **Pareto Analysis:**

   To determine the contribution of clients to the total AUM, pareto analysis provides an insightful distribution of the AUM with respect to clients. For this, the data was sorted based on client AUM in descending order. Then, cumulative AUM was determined accordingly. The plotting of client AUM data was done using bar chart and cumulative AUM was plotted using a line chart with a secondary axis holding the percentage of the total AUM. This analysis aims at determining the clients that are significant to the revenue of the organisation currently and hence services to them should be catered accordingly.

   Since this method particularly takes into consideration only the current AUM of client and organisation, identifying high-value customers from this would remain ambiguous. This happens because AUM of a client depends on multiple factors like investment duration and new/periodic investments through SIP or lumpsum purchases. Though pareto analysis remains crucial for avoiding major losses in the AUM, limiting our goal of finding high-value clients using this method is not sufficient.

   To verify the above argument, the time of association of a client with the organisation as a metric should also be considered to be analysed. Therefore, calculation of the moving average of the "Months" column on the already sorted data and plotting it accordingly is the next step.

2. **Monthly AUM:**

   As observed from the previous method, a new metric that attributes better judgements about customers with respect to their investments is required. The metrics that were considered for investment analysis were "SIP", "AUM", redemptions, and the time of association with the organisation ("Months"). Since investments made by the clients are goal based and sometimes redemptions are made when those goals are fulfilled or in light of an emergency, current AUM is not sufficient and redemptions should also be

accounted. So, a feature called "Monthly AUM" was determined from "AUM", "Redemption amount" and "Months", where:

Monthly AUM = (AUM + Redemption amount) / Months

This includes SIP and lumpsum investments along with the redemptions that have been made, and normalised with respect to time. This metric will be thoroughly used as a target variable for modelling and segmenting the investments made based on client profiles.

3. **Customer Segmentation**

<u>According to client profile</u>

Segmentation of clients is an important process in the acquisition and retention strategies. The aim of this analysis will be to group the client base based on personal features (other than numerical/investment features).

The segmentation process involves clustering of the clients without any target variable. This type of modelling comes under unsupervised learning. The choice of algorithm for the baseline clustering would be K-Means clustering. This algorithm clusters the data using Euclidean distance between the data points and then forming cluster centroids and therefore grouping similar data points together. These clusters would then be associated with each client in the data and to determine the significance of these clusters, investment features will be analysed based on these clusters. If clear distinction between the investment metrics is found among these clusters, it will mean that the clusters can be considered to be a good segmentation strategy.

The parameters on which these clusters are to be formed are:

[Monthly Income, Risk Appetite, Network Degree, Gender, Mode of Income, Marriage Status, Responsibility, Earning Individual, Age]

Method flow:

Scaling the data → Determine number of clusters (k) → K-Means clustering

(Standard scale) (Elbow method) (Investment analysis)

4. **Network degree analysis**

After segmentation of clients using clustering techniques which proves to be efficient in grouping individuals based on their profiles, and henceforth forms a general overview of various natures of clients. Judging investment potential, mindset and strategy for new and existing clients becomes more straight forward.

Network degree analysis adds a layer of insight for an acquisition strategy. As segmentation answers the question of whom to target, network degree aims at providing an answer for how to reach out.

Comparison of clients based on network degree (1st/2nd/3rd) and overall averages against investment metrics is performed using pivot table. This gives a rough idea on the mindset of investors with respect to intermediaries like M K Financial Services.

TIME_SERIES

1. **AUM and Client Growth Modelling**

   This process helps to model the AUM which will form the base to forecast the AUM growth as well as client growth. The quarterly data of AUM is plotted and then a trendline is determined. The R-squared value is checked for the trendline to judge the goodness of fit.

   A linear model is first incorporated and then a polynomial function with degree two is checked. Whichever has a better R-squared value will be considered for forecasting. The under-performance and over performance are analysed with respect to the time. These errors have the potential to explain the influence of external market factors i.e. the return on investments and also of the number of clients.

   The number of clients with respect to time is also plotted to determine the past performance of the organisation with respect to client acquisition.

2. **Compound Annual Growth Rate (CAGR)**

   To calculate CAGR for AUM, the first two years won't be considered because it skews the growth trend because of high inflows when the business was in its initial phase.

   CAGR = ((AUM final/AUM initial)^(1/years)) – 1

   The client growth rate is of a constant type which can be determined by the linear equation of the type

   Number of clients = (growth rate*years) + constant

   The return on investment generally considered for mutual funds is the same as index growth which is 12% annually. But since most of the investments made by the organisation are in equity class and because of the alpha provided by the fund manager

and M K Financial Services, the return on investment considered for analysis will be 15%.

3. **Target Growth Rate**

Since it was observed that AUM is a function of both client growth and previous AUM, the model can be viewed as:

$AUM(t+1) = AUM(t)\times(1+r) + \text{New Inflow}(t+1)$

where:

r is the effective return rate on existing investments.

New Inflow(t+1) comprises contributions from new clients and/or increased contributions from existing clients.

To achieve a target growth rate T:

$AUM(t+1) = AUM(t)\times(1+T)$

By equating the above two expressions:

$AUM(t)\times(1+T) = AUM(t)\times(1+r) + \text{New Inflow}(t+1)$

Then solving for the new inflow required:

$\text{New Inflow}(t+1) = AUM(t)\times(T-r)$

This tells that to reach a target growth rate T, the organisation must generate additional inflows equal to the difference between the target growth and the return growth, multiplied by the current AUM. These new inflows then can be further be divided into targets for new clients and additional investments by existing clients.

To achieve these AUM targets, we consider inflows only through SIP. The current SIP book of the organisation is around Rs. 16.5 lakhs. The average SIP for a client is Rs. 6,200. The average "Monthly AUM" for a client is Rs. 1.2 lakhs.

Using this information, a deficit is calculated of the required new inflows to meet target AUM and the new inflows through already existing SIPs. This deficit will then be divided according to the average SIP per client totalled to yearly amount to get the number of new clients required to meet the target AUM for that year.

4. **Regression analysis to predict number of clients to meet the forecasted AUM**

Regression was performed on the training data with the number of clients as explanatory variable and the AUM as target variable and a polynomial fit was found with an R-squared value of 0.98. The polynomial equation with degree two which

seems a good fit was then used to solve for the number of clients when the forecasted AUM values were substituted.

These values will be considered for target setting of client acquisitions.

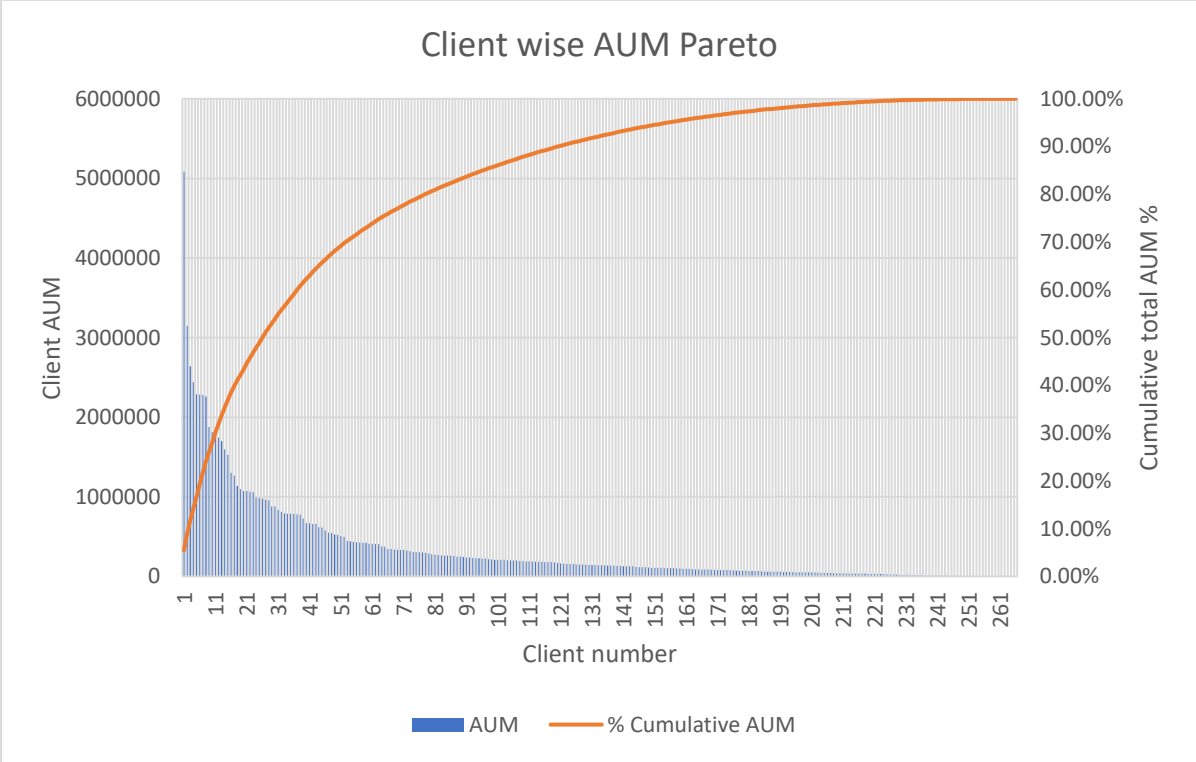
3 Results and Findings

CLIENT_MASTER

1. **Pareto Analysis:**

Figure-1 shows the presence of current high-value clients on which the organisation revenue depends on. It was seen that roughly top 25% of the clients, that is around 70 clients are responsible for more than 75% of the total revenue.

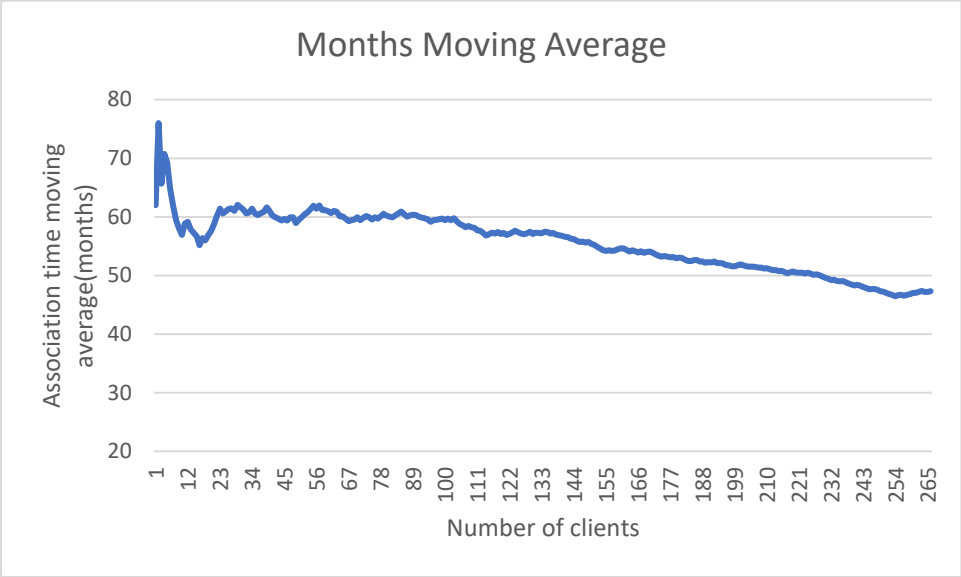
Figure-1



As seen in Figure-2, the moving average decreases with clients when sorted in descending order with respect to their AUM. The implication of this is that generally the clients who remain low in the pecking order (AUM), are relatively new clients. So, client AUM is not sufficient to categorise customers based on the value they hold for the organisation.

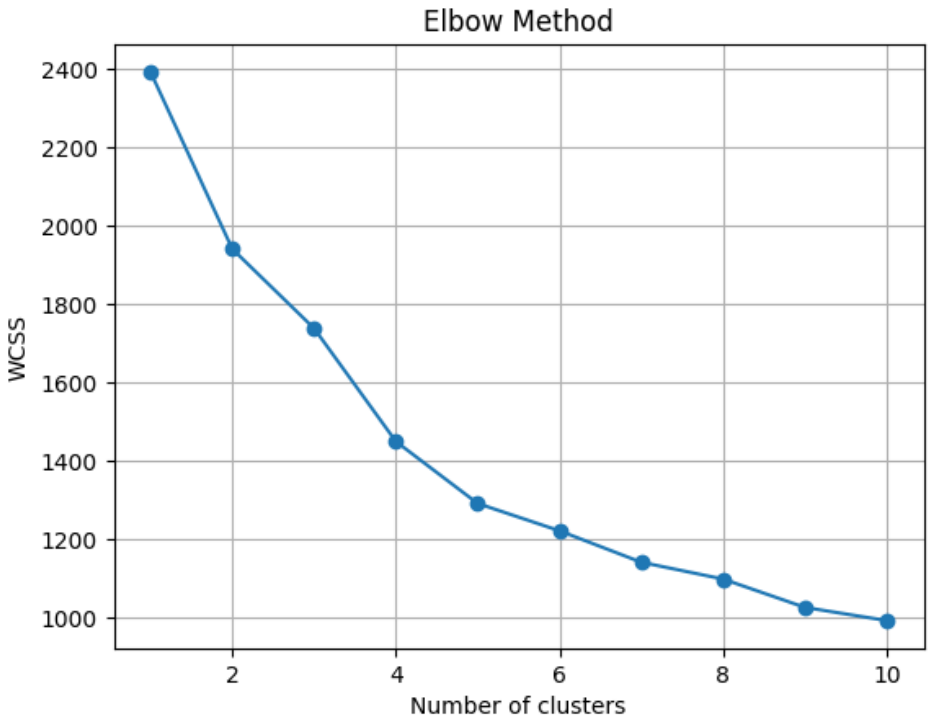
Determination of a new derived metric that considers association time along with the AUM is required to judge customer value more precisely.

Figure-2



2. Customer Segmentation

Figure-3



The optimal number of clusters(k) which will be considered for clustering the data points will be 4 since the decrease in WCSS(Within Cluster Sum of Squares) starts to level off after k=4. (Figure-3)

Figure-4

Number of clients in each cluster:

0	47
1	65
2	44
3	110

Cluster-wise Client Profile Summary:

Cluster	**Monthly Income**	**Risk apetite**	**Network degree**	**Gender (M)**	**Mode of income (Salaried)**	**Marriage Status (Single)**	**Responsibility (Yes)**	**Earning Individual (Self)**	**Age**
0	0.91	2.00	1.45	0.60	0.30	0.00	0.83	1.00	43.57
1	0.08	1.06	1.48	0.02	0.40	0.00	0.68	0.48	47.03
2	0.36	1.34	1.14	0.75	0.41	1.00	0.27	0.80	24.39
3	0.75	0.99	1.47	0.85	0.37	0.02	1.00	1.00	44.30

To verify the goodness of these clusters to distinguish these clients based on their investments, comparison of the central tendencies of numerical features (Monthly AUM) between clusters was performed.

Figure-5

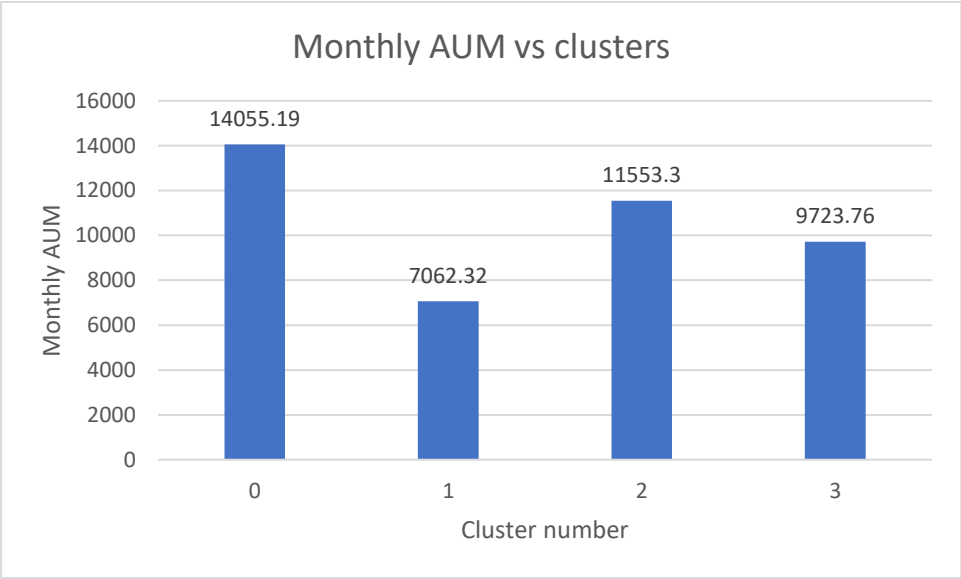
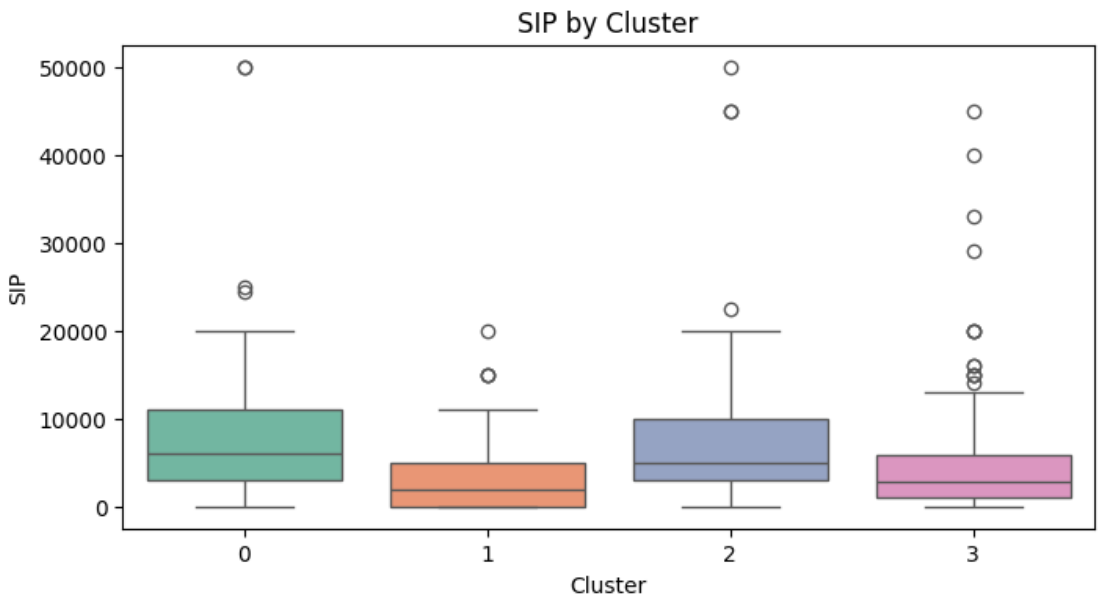


Figure-6

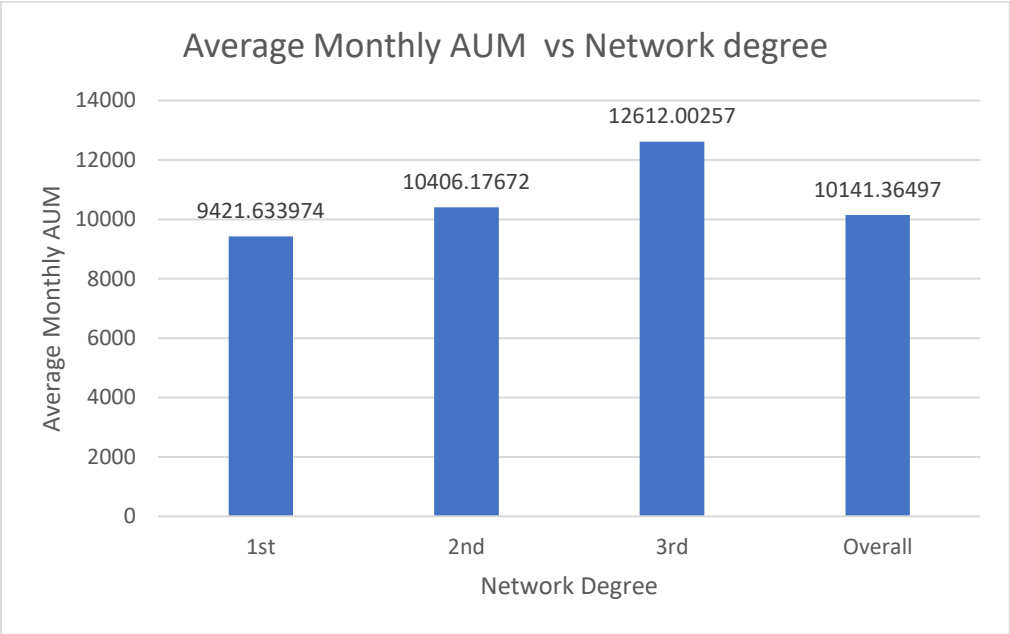


Cluster	AUM	Monthly AUM	SIP	Redemption amount	Redemption count
0	595833.06	14055.19	9255.32	109086.71	2.79
1	192116.12	7062.32	3415.38	97951.38	3.97
2	333163.92	11553.3	9022.73	41227.43	1.07
3	344269.94	9723.76	5459.09	127127.39	1.9

The clusters based on client profiles are able to distinguish the investment features decently. After determining the cluster significance and meaning, they can form a baseline strategy for further acquisitions and determine investment profiles based on individual personal details.

3. **Network degree analysis**

Figure-7



As seen from Figure-7, monthly AUM of clients tend to increase with increase in their network degree connection with the organisation.

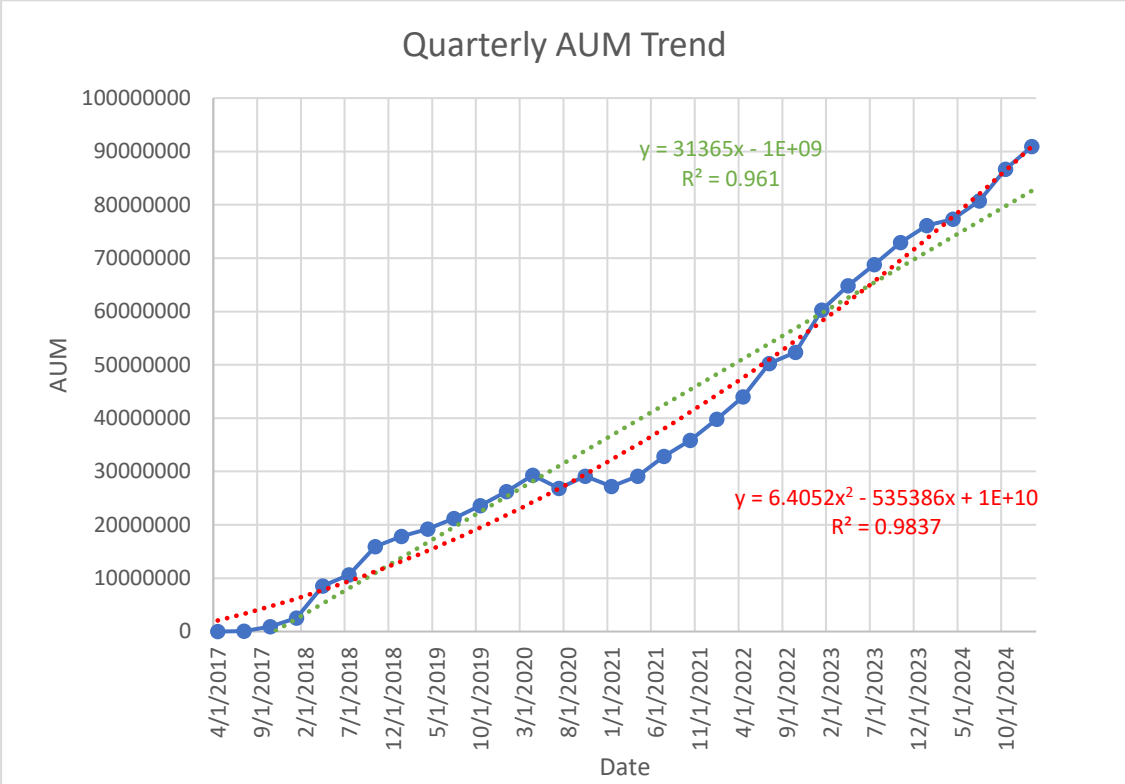
More than half of the clientele are directly connected to the organisation, around 35% clients are referred connections and 11% belong to the 3rd network degree.

The analysis here contradicts the existing acquisition strategy. It shows that the organisation relied on their direct connections like friends and family as clients and still their average monthly AUM contribution remains less than of the referred/out of network clients.

TIME_SERIES

1. **AUM and Client Growth Modelling**

Figure-8



From the above comparison of R-squared value, it is evident that the AUM growth is a polynomial function with degree two. Though, the R-squared value is only marginally more for the polynomial than the linear fit, we still conclude that AUM growth is polynomial. The reason for this is intuitive given the nature of returns on investments. AUM is intuitively a function of the total investments and the returns realised on the same. The total and new investments remain a subject to be analysed further. Since the returns are compounded on a daily basis, the intuitive approach for returns would be exponential.

Figure-9

The client growth is more or less linear. From the equation on the Figure-9, it can be said that historically around 27 new clients are acquired annually.

2. CAGR

The observed average CAGR for AUM is 31.7% over a period of six years.

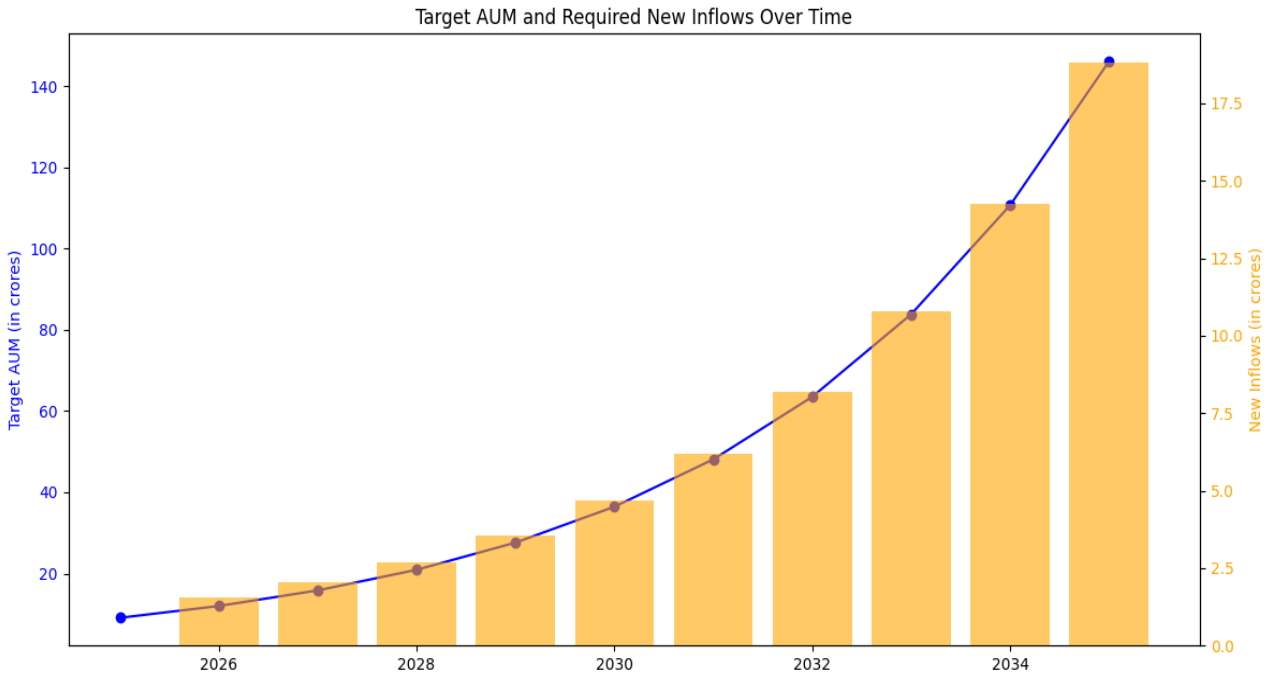
The average return on investment being 15%.

And the average number of new clients the organisation acquires is 27.

3. Target Growth Rate

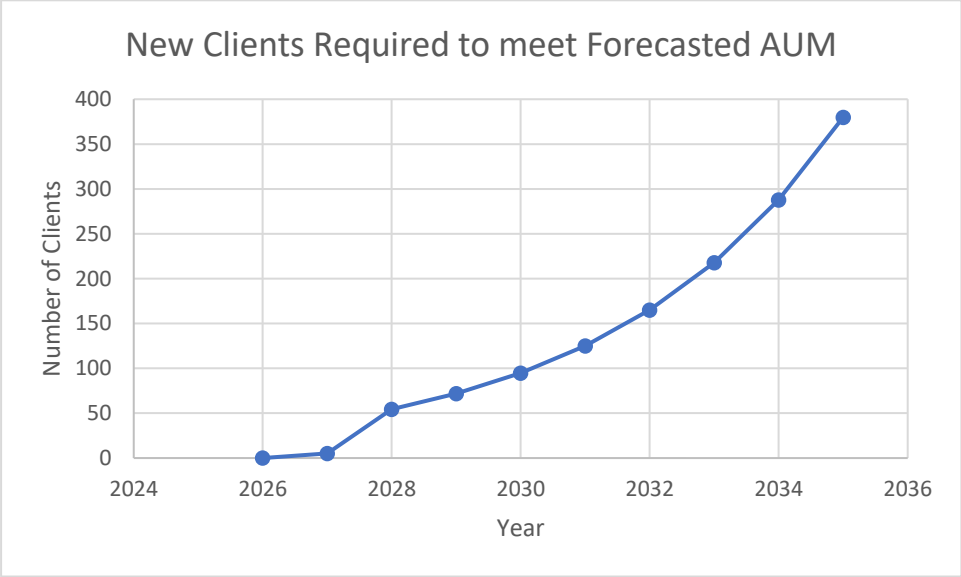
To maintain the current CAGR of approximately 32%, the new inflows for year 2025 on top of the previous year AUM of Rs. 9,09,12,389 is around 1.5 crores which covers the deficit of 17% to reach the target AUM, if return on investments considered are 15%. This return on investment is subject to market conditions and is not fixed.

Figure-10



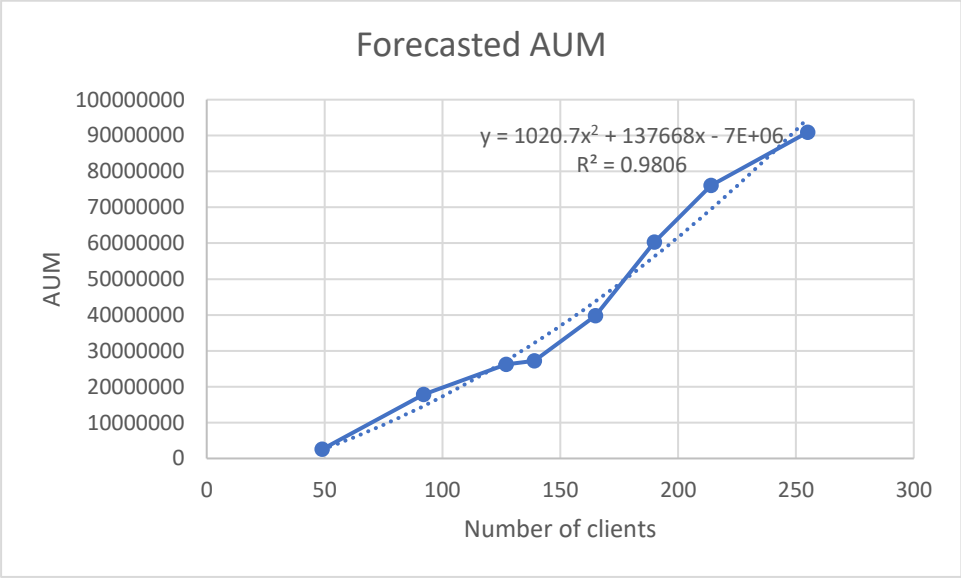
To maintain the growth rate of AUM for the next 10 years, the number of new clients required is plotted on Figure-11. This analysis excludes lumpsum investments and also the possibility of existing clients increasing their SIP amount in the future.

Figure-11



4. Regression analysis to predict number of clients to meet the forecasted AUM

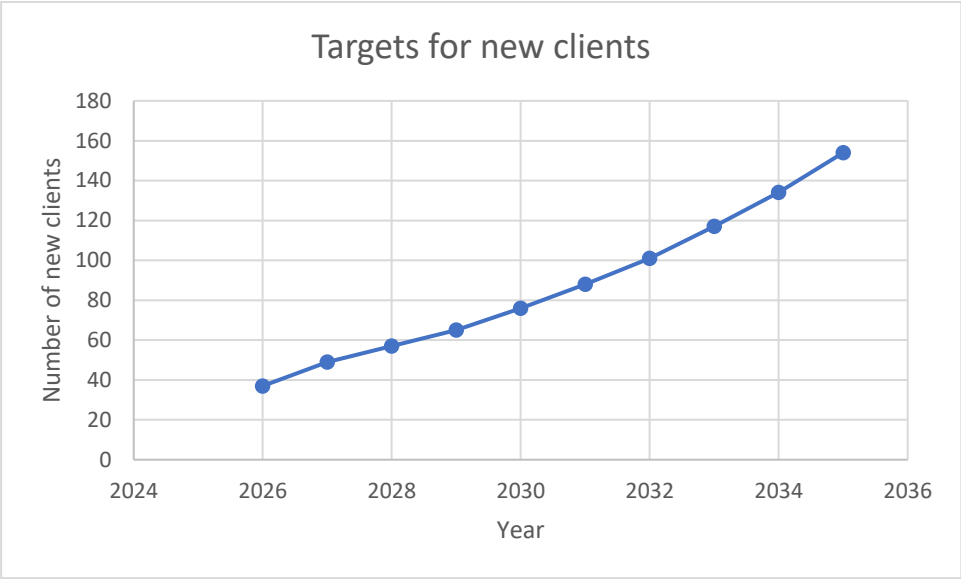
Figure-12



This bivariate analysis (Figure-12) shows the heavy dependency of AUM on number of clients. The trend equation is then used to solve inversely for the number of clients given the forecasted AUM. These target values of number of clients at then end of each year is plotted in Figure-13. It is evident form the chart that client growth should not be linear if the target AUM is to be met.

On average, around 80 clients should be acquired annually. This number will act as the sales target for every year.

Figure-13



4 Interpretation of Results and Recommendations

Client Segmentation and Acquisition Strategy (objective -1)

1. In order to prevent unforeseen and significant losses in revenue, the organisation should provide catered services to the top 70-80 AUM contributors. Strategies like portfolio rebalancing depending on the equity market scenario and frequent communication should be adapted in order to proactively engage and henceforth retain these clients. With respect to the current revenue generation, these top investors remain critical.
2. Segmentation of clients based on their personal profiles would serve an instrumental purpose in determining the investment potential and horizons of a client.
   After performing K-Means clustering on client profile data, four clusters were obtained. The significance of those clusters and subsequent investment patterns are as follows:
   - **Cluster-0:**
     High income individuals with a higher risk appetite. These belong to a relatively older age group, are married and have responsibilities. These are self-earning individuals and most of them run businesses.

The average "Monthly AUM" of these clients (Rs. 14,055) is highest among all the clusters. Clients have made decent redemptions. Still, the average "AUM" in this cluster is impressive.

It can be stated that clients belonging to this cluster engage actively with mutual funds as an investment instrument.

- **Cluster-1:**

  These are low income, conservative and older clients. Majority of them are married women who are aliases for investments.

  The average "Monthly AUM" of these clients (Rs. 7,062) is the lowest among all the clusters. The same goes for the average "SIP" and "AUM" in this cluster.

- **Cluster-2:**

  The clients who belong in this cluster are significantly younger than of the other clusters. These are salaried with moderate income and risk. They have less responsibilities and are single.

  These cluster characteristics reasonably explain the good average "SIP" and "Monthly AUM" (Rs. 11,553) of the clients. The redemption patterns show that these clients have not made many redemptions.

  Overall, this cluster shows promising clientele with a long-term investment/appreciation mindset.

- **Cluster-3:**

  This is the most populated cluster and is similar to Cluster-0 with a little more conservative risk profile and slightly lower monthly income. All the individuals in these cluster have responsibilities.

  The investment features seem to be significantly less than Cluster-0. The average "Monthly AUM" (Rs. 9,723) is a decent and moderate investment strategy adapted by these clients.

  This cluster has middle-aged, working, middle-class individuals.

3. As per the network degree analysis of clients, it can be deduced that average monthly investments increase for clients who were referred by some existing client. This can imply that individuals who are themselves willing to invest through mutual funds look for references to some organisation as M K Financial Services for 'hand-holding' to build their portfolio over time.

The reason for a lower "Monthly AUM" for direct connections can be an obligatory mindset amongst the individuals when approached by the organization. The purpose of investment changes in this case from actively believing in capital appreciation to investing for the sake of it because of close inter-personal relations.

4. A good acquisition strategy that can be formulated using the above results and interpretations is a reference-based approach. The type of individuals that can be targeted and have the potential to be easily converted are young salaried individuals who have less responsibilities and are at their early-career stages. References can be used to further expand the active network of business running middle-aged individuals.

Growth Forecast and Sales Target Determination

1. The market historically follows a cyclical growth trend. By that it means, a single market cycle consists of crest and trough generally. Sometimes, when unforeseen incidents globally can also shift the nature of the market movement.

   From the AUM growth trend, it is observable that the trend is being out-performed and under-performed at various intervals over time. From early 2020 to late 2021 the organisation AUM slightly underperforms the growth trend. This can be summed up as a direct result of a market downfall during the Covid pandemic. The market faced a significant correction during the early 2020 and since then it took more than a year to recover to the previous levels which can be very well verified from the quarterly AUM. Since 2022, the market has been going through a bull run with minor corrections which can again be verified from the quarterly AUM out-performing the growth trend.

   This shows that though the AUM over long term follows a growth trend, but minor fluctuations happen as a result of micro and macro-economic factors. This can affect the revenue of the organisation temporarily.

   Therefore, certain asset classes like hybrid funds involving other assets than equity like gold, corporate bonds and debt can balance these minor short comings related to equities. Therefore, bigger portfolios can be diversified using these hybrid funds to minimise the volatility of client AUM and in turn the revenue of the organisation.

2. The polynomial (degree=2) nature of AUM over time suggests that AUM depends on return on investments and capital. This capital is further divided in terms of existing

and new clientele. It also suggests the impact of periodic investments through SIPs contributing to the polynomial nature.

3. The CAGR is therefore determined. It remains ambiguous as to how much part of the CAGR is a result of new clients. Hence the target growth rate of the AUM is considered as the average CAGR.

   The number of new clients required annually was therefore determined intuitively using expected new inflows. But the results remained questionable due to the assumptions of new inflows only though SIPs and not considering additional investments through existing clients.

   Therefore, a regression strategy with one variable made more sense.

4. The target for the organisation in terms of client acquisition should be around 80 clients every year. This will help achieve the forecasted AUM of the organisation and might even out-perform it.